Chenan Wen

Ph.D. Candidate, Computer Engineering — Trustworthy Systems, Formal Verification & Secure/Private AI

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Summary

Ph.D. candidate in Computer Engineering at Purdue working at the intersection of **security**, **formal verification**, **programming languages**, and **AI/ML** to build systems with strong, verifiable guarantees. First-author research spans LLM-assisted formal verification of security policies (**accepted to ACM SIGCOMM 2026**), SMT-based compiler/DSL optimization, and RISC-V computer architecture — giving me security/privacy depth co-designed across the full hardware/software stack. Seeking a Security & Privacy research internship building trustworthy AI systems with strong, **verifiable security, privacy, and safety guarantees**.

Education

Purdue University, Ph.D. in Computer Engineering – West Lafayette, IN — Jan 2022 – present

- Focus: security, formal methods, programming languages, and applied AI / LLMs for trustworthy systems

Northwestern University, M.S. in Computer Engineering – Evanston, IL — Sept 2019 – Aug 2022

Beihang University, B.S. in Electrical Engineering & Automation – Beijing, China — Sept 2015 – Aug 2019

Research Experience

Graduate Research Assistant — Secure AI & Formal Verification, Purdue University – West Lafayette, IN — June 2024 – present

- Built **ElastiSpec** (accepted, **ACM SIGCOMM 2026**), a framework that formally audits enterprise firewall configurations against natural-language security policies, detecting misconfigurations that permit non-compliant traffic.
- Designed a security-policy DSL and a formal auditor using **angelic verification** (extending Batfish) that certifies compliance under partial specifications and reports verifiable guarantees with conjectures for human validation.
- Engineered a multi-step **LLM agent** (RAG / LangChain) to translate ambiguous vendor documents into the formal DSL, raising extraction accuracy from 61% to 96% while formally checking every AI-produced artifact for logical consistency — analyzing and mitigating the risks of placing AI in high-assurance security loops.
- Evaluated on real-world configurations across **70+ enterprise firewalls**, uncovering 15 security misconfigurations (legacy ACLs, flipped traffic directions, incorrect protocol–port mappings).

Graduate Research Assistant — Programming Languages & Compilers, Purdue University – West Lafayette, IN — Sept 2023 – June 2024

- Developed **P4CGO** (ACM SIGCOMM FMANO 2024), a DSL and compiler that synthesizes optimized P4 programs and match-action tables for programmable switches from high-level policy sketches.
- Built a formal table-compression engine in Java using the **Z3 SMT solver** to split and merge policy tables, achieving ∼64% average compression on legacy firewall rulesets while preserving semantics.
- Implemented a reverse-engineering tool that lifts low-level match-action tables back into the high-level policy language.

Graduate Researcher — Computer Architecture, Northwestern University – Evanston, IL — July 2020 – July 2021

- Synthesized the open-source **RISC-V BOOM** out-of-order multicore SoC to a gate-level netlist with **Synopsys Design Compiler** ran dynamic timing analysis of critical paths across microbenchmarks.
- Tuned pipeline depth to improve timing slack across multiple process/voltage/temperature (PVT) corners.

Selected Publications

ElastiSpec: Formalizing Enterprise Firewall Management with Informal and Elastic Specifications — July 2026
Chenan Wen, Yizhan Qing, Curt Jansen, Xiaokang Qiu, Sanjay Rao
(ACM SIGCOMM 2026 (accepted))

P4CGO: Control Plane Guided P4 Program Optimization — Aug 2024
Chenan Wen, Zhuocong Li, Syed Usman Jafri, Xiaokang Qiu, Sanjay Rao
(ACM SIGCOMM Workshop on FMANO 2024)

Skills

Programming: Python, C++, C, Java

Formal Methods & Verification: Z3 / SMT, MiniSAT, Batfish, program synthesis, ANTLR4

Trustworthy AI / ML: LLM agents, safe/verifiable use of LLMs, Retrieval-Augmented Generation (LangChain)

Security, Privacy & Systems: Firewall / ACL & access-control analysis, network verification, P4 / SDN, RISC-V computer architecture

Languages: Mandarin (native), English (fluent)